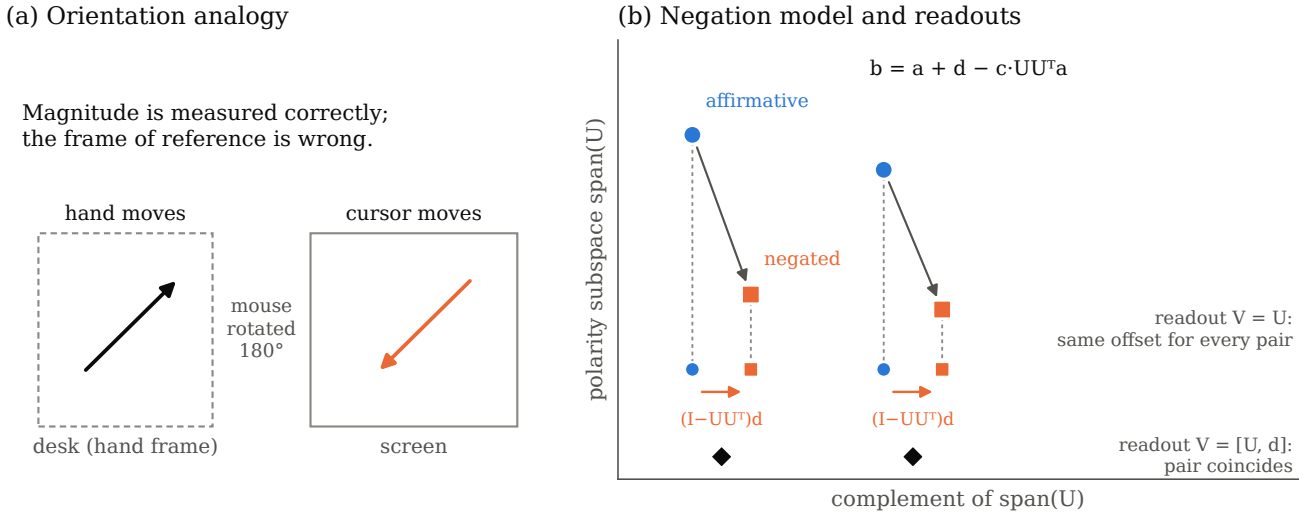
(a) Orientation analogy
Magnitude is measured correctly;
the frame of reference is wrong.
hand moves
cursor moves
mouse
rotated
180°
desk (hand frame)
screen
(b) Negation model and readouts
b = a + d − c·UUᵀa
affirmative
negated
polarity subspace span(U)
(I−UUᵀ)d
(I−UUᵀ)d
readout V = U:
same offset for every pair
readout V = [U, d]:
pair coincides
complement of span(U)